


# University of Asia Pacific

**Department of Computer Science and Engineering**

## Lab Report 3

# CMOS XOR Gate, Full Adder and 2×1 Multiplexer

**Assignment: Combined CMOS XOR, Full Adder & 2×1 MUX Design, Simulation, and Symbol Creation**

Course name: VLSI Design Lab

Course Code: CSE 450

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Section: E1

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# 1. Pre-Lab Work

## 1.1 Objective

The objective of this lab is to study the structure and operation of the two-input static CMOS XOR gate, the one-bit Full Adder and the 2×1 multiplexer (MUX), determine suitable transistor sizing, create the circuits in Cadence Virtuoso using the gpdk045 library, verify their logic function through transient simulation, and create reusable schematic symbols.

## 1.2 MOSFET Fundamentals

A MOSFET is a voltage-controlled switching device with four terminals: Gate, Drain, Source, and Body. CMOS logic uses complementary NMOS and PMOS devices.

- NMOS (n-channel): turns ON when VGS exceeds the NMOS threshold and provides a conducting path toward VSS. It passes a strong logic 0 but a weak logic 1.
- PMOS (p-channel): turns ON when VSG exceeds |VTp| and provides a conducting path toward VDD. It passes a strong logic 1 but a weak logic 0.
- In static CMOS, PMOS devices form the pull-up network (PUN) and NMOS devices form the pull-down network (PDN).
- Transmission gate (TG): an NMOS and a PMOS connected in parallel with complementary gate signals. It passes both logic levels without threshold loss and is used here for the 2×1 MUX.

## 1.3 CMOS XOR Gate Structure and Operation

A two-input XOR gate implements $Y = A \oplus B = A'B + AB'$. The XOR gate in this lab is a static complementary CMOS gate built as the complement of XNOR: $Y = \neg(A \cdot B + A' \cdot B')$. Two inverters generate $A'$ and $B'$. The pull-down network contains two series NMOS branches ($A \cdot B$ and $A' \cdot B'$) connected in parallel, and the pull-up network is its dual: two parallel PMOS pairs connected in series. The complete gate uses 12 transistors (8 in the main network and 4 in the two input inverters).

- A = 0, B = 0: the $A' \cdot B'$ NMOS branch conducts → Y = 0.
- A = 0, B = 1: both NMOS branches are broken and the PMOS network conducts → Y = 1.
- A = 1, B = 0: both NMOS branches are broken and the PMOS network conducts → Y = 1.
- A = 1, B = 1: the $A \cdot B$ NMOS branch conducts → Y = 0.

## 1.4 CMOS Full Adder Structure and Operation

A one-bit Full Adder adds three input bits A, B and carry-in Cin and produces Sum and carry-out Cout:

- $Sum = A \oplus B \oplus Cin$
- $Cout = A \cdot B + A \cdot Cin + B \cdot Cin$ (equivalent to $A \cdot B + Cin \cdot (A \oplus B)$)

The Full Adder is built at gate level from the cells created earlier in the lab. The Sum path uses two cascaded XOR gates. The carry path forms the three product terms $A \cdot B$, $A \cdot Cin$ and $B \cdot Cin$ with three NAND gates, each followed by an inverter, and then combines them with two OR stages, each realized as a NOR gate followed by an inverter. The complete adder therefore contains 2 XOR, 3 NAND, 2 NOR and 5 inverter cells (54 transistors).

## 1.5 CMOS 2×1 Multiplexer Structure and Operation

A 2×1 MUX selects one of two data inputs (I0, I1) according to the select line S (pin Set in the schematic): $Y = S' \cdot I0 + S \cdot I1$. The MUX in this lab uses two transmission gates and one inverter (6 transistors). TG1 passes I0 when S = 0 and TG2 passes I1 when S = 1; the inverter generates $S'$ for the complementary gate control.

- S = 0: TG1 is ON, TG2 is OFF → Y = I0.
- S = 1: TG1 is OFF, TG2 is ON → Y = I1.

## 1.6 Design and Sizing

For a first-order CMOS sizing approach, RON is approximately proportional to L/W, so increasing transistor width reduces its effective ON resistance. Ideally, transistors in a series stack and PMOS devices (lower hole mobility) would be made wider than a single NMOS reference device.

In this lab the circuits were designed with the gpdk045 library using the nmos1v and pmos1v devices (models g45n1svt and g45p1svt). Every transistor in the XOR gate and in the MUX uses the same size, W = 120 nm, L = 45 nm and m = 1. The Full Adder is built entirely from previously designed gate cells, so no new transistor sizes were introduced. Using equal widths keeps the design simple and consistent, at the cost of weaker series stacks and unequal rise and fall times, which is discussed in Section 5.

| Circuit | Device | Model | W | L | m | Topology |
|---|---|---|---|---|---|---|
| XOR | PDN NMOS NM0–NM3 (×4) | g45n1svt | 120 nm | 45 nm | 1 | Two series branches in parallel |
| XOR | PUN PMOS PM0–PM3 (×4) | g45p1svt | 120 nm | 45 nm | 1 | Two parallel pairs in series |
| XOR | Input inverter cells (×2) | Inverter cell | As in cell | As in cell | 1 | Generate A′ and B′ |
| Full Adder | XOR ×2, NAND ×3, NOR ×2, INV ×5 | Sub-cells | As in cells | As in cells | 1 | Hierarchical gate level |
| 2×1 MUX | TG NMOS NM1, NM2 | g45n1svt | 120 nm | 45 nm | 1 | Transmission gate |
| 2×1 MUX | TG PMOS PM1, PM2 | g45p1svt | 120 nm | 45 nm | 1 | Transmission gate |
| 2×1 MUX | Inverter NM0 / PM0 | g45n1svt / g45p1svt | 120 nm | 45 nm | 1 | Select inverter |

## 1.7 Expected Truth Tables (Pre-Lab Prediction)

**XOR: Y = A ⊕B**

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

**Full Adder: Sum = A ⊕B ⊕Cin, Cout = AB + ACin + BCin**

| A | B | Cin | Sum | Cout |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 |

**2×1 MUX: Y = S′·I0 + S·I1**

| S | I0 | I1 | Y |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 |

## 1.8 Pre-Lab Checklist

- Reviewed NMOS and PMOS switching behavior, CMOS pull-up/pull-down networks, and transmission gates.
- Derived the XOR, Full Adder and MUX Boolean expressions and truth tables.
- Identified the series/parallel transistor topology of the XOR gate and the transmission-gate structure of the MUX.
- Planned the gate-level hierarchy of the Full Adder using the XOR, NAND, NOR and inverter symbols.
- Planned the Cadence schematic connectivity and supply connections.
- Planned the transient test benches using analogLib vpulse, vdc and gnd.
- Predicted the logic and transient switching behavior before simulation.

# 2. Truth Table and Circuit Description

## 2.1 XOR Truth Table

| A (Input) | B (Input) | Y = A ⊕B (Output) |
|---|---|---|
| 0 (LOW) | 0 (LOW) | 0 (LOW) |
| 0 (LOW) | 1 (HIGH) | 1 (HIGH) |
| 1 (HIGH) | 0 (LOW) | 1 (HIGH) |
| 1 (HIGH) | 1 (HIGH) | 0 (LOW) |

## 2.2 XOR Circuit Description

The implemented XOR circuit is a 12-transistor static CMOS gate. Two inverters produce A′ and B′. The output node (pin Vout) is connected to Vdd through the PMOS network and to Vss through the NMOS network.

- PDN: two series NMOS branches in parallel between the output and Vss, one branch controlled by A and B and the other by A′ and B′.
- PUN: two parallel PMOS pairs in series between Vdd and the output, one pair controlled by A and B and the other by A′ and B′.
- PMOS bulks are tied to Vdd and NMOS bulks are tied to Vss.
- The output is LOW when A = B (a pull-down path exists) and HIGH when A ≠ B (a pull-up path exists).

## 2.3 Full Adder Truth Table

| A | B | Cin | Sum | Cout |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 |

## 2.4 Full Adder Circuit Description

The implemented Full Adder is a hierarchical gate-level design that instantiates the symbols of the verified XOR, NAND, NOR and inverter cells (54 transistors in total).

- Sum path: XOR1 computes P = A ⊕B and XOR2 computes Sum = P ⊕Cin.
- Product terms: NAND(A,B), NAND(A,Cin) and NAND(B,Cin), each followed by an inverter, give A·B, A·Cin and B·Cin.
- Carry path: NOR + inverter combines A·B and A·Cin, and a second NOR + inverter combines that result with B·Cin, giving Cout = A·B + A·Cin + B·Cin.
- Vdd and Vss pins of every instantiated cell are connected to the common supply rails.

## 2.5 2×1 MUX Truth Table

| S | I0 | I1 | Y (Output) |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 |

## 2.6 2×1 MUX Circuit Description

The implemented MUX is a transmission-gate multiplexer with two transmission gates and one inverter (6 transistors).

- The inverter (NM0, PM0) generates S′ from the select input S (pin Set).
- TG1 (NM1, PM1) connects I0 to Y when S = 0.
- TG2 (NM2, PM2) connects I1 to Y when S = 1.
- The outputs of TG1 and TG2 are joined at node Y; only one transmission gate conducts at any time.

# 3. Circuit Schematics and Simulation Results

## 3.1 Simulation Setup

All circuits were designed with the gpdk045 library in Cadence Virtuoso and simulated with Spectre. The test benches use components from analogLib: a vdc source for the supply, vpulse sources for the inputs, and a gnd reference.

| Parameter | Value |
|---|---|
| Technology / devices | gpdk045: nmos1v (g45n1svt) and pmos1v (g45p1svt) |
| Supply (vdc) | VDD = 1.8 V |
| Input source (vpulse) | V1 = 0 V, V2 = 1.8 V, delay = 0, rise time = 1 ns, fall time = 1 ns |
| Input periods / pulse widths | 20 ns / 10 ns, 40 ns / 20 ns, 60 ns / 30 ns |
| Transistor size | W = 120 nm, L = 45 nm, m = 1 |
| Analysis | Transient, stop time 120ns |

The three pulse trains have periods 20 ns, 40 ns and 60 ns with pulse widths equal to half the period, so that the inputs step through different input combinations without needing a separate pattern generator. Assignment of the sources to the circuit inputs:

| Circuit | Source | Connected to | Period | Pulse width |
|---|---|---|---|---|
| XOR | V0 | A | 20 ns | 10 ns |
| XOR | V1 | B | 40 ns | 20 ns |
| Full Adder | V0 | A | 20 ns | 10 ns |
| Full Adder | V1 | B | 40 ns | 20 ns |
| Full Adder | V2 | Cin | 60 ns | 30 ns |
| 2×1 MUX | V0 | I0 | 20 ns | 10 ns |
| 2×1 MUX | V1 | I1 | 40 ns | 20 ns |
| 2×1 MUX | V2 | Set (S) | 60 ns | 30 ns |

## 3.2 XOR Schematic Capture

The XOR gate was captured in Cadence Virtuoso Schematic Editor using the nmos1v and pmos1v devices and the 12-transistor complementary topology described in Section 2.2. The two inverters at the inputs generate A′ and B′.

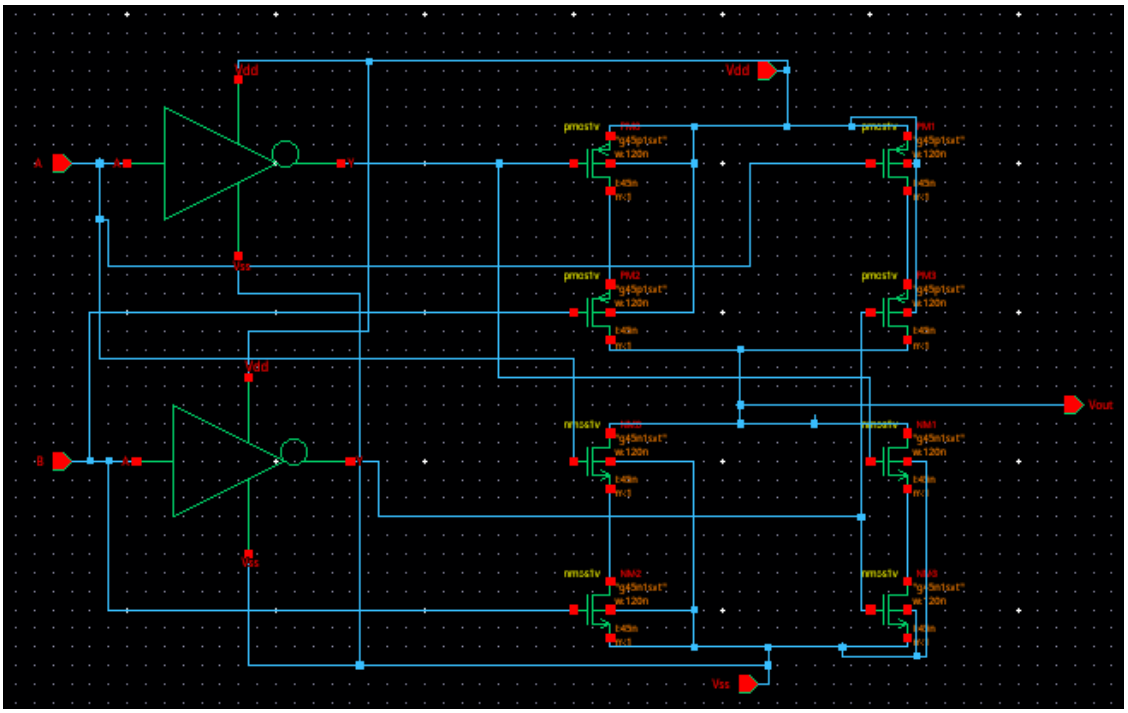



**Figure 1. CMOS XOR gate schematic captured in Cadence Virtuoso.**

## 3.3 XOR Test Bench and Transient Response

The XOR symbol was placed in a test bench with the vdc supply and two vpulse inputs (Figure 2). The output is expected to be HIGH only while exactly one of the inputs is HIGH. Finite rise/fall times appear because practical MOSFETs have non-zero on-resistance and the output node has capacitance.

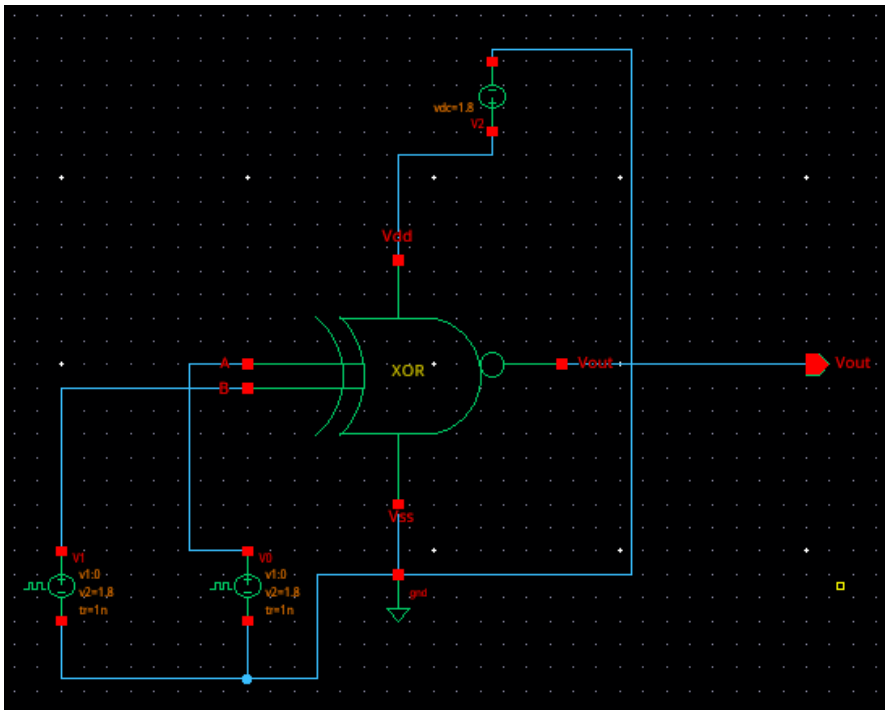



**Figure 2. Test bench for the XOR transient simulation.**

In the transient plot of Figure 3, net1 is input A (20 ns period) and net2 is input B (40 ns period).




**Figure 3. Transient response of inputs A (net1), B (net2) and output Vout (XOR).**

| Time window (ns) | A | B | Expected Y | Observed Y |
|---|---|---|---|---|
| 2 – 12 | 1 | 1 | 0 | 0 (LOW) |
| 12 – 22 | 0 | 1 | 1 | 1 (HIGH) |
| 22 – 32 | 1 | 0 | 1 | 1 (HIGH) |
| 32 – 42 | 0 | 0 | 0 | 0 (LOW) |
| 42 – 52 | 1 | 1 | 0 | 0 (LOW) |

## 3.4 XOR Device Parameter Summary

| Parameter | PMOS PM0–PM3 | NMOS NM0–NM3 |
|---|---|---|
| Device cell | pmos1v | nmos1v |
| Device model | g45p1svt | g45n1svt |
| Width (W) | 120 nm | 120 nm |
| Length (L) | 45 nm | 45 nm |
| Multiplier (m) | 1 | 1 |
| Network | PUN (parallel pairs in series) | PDN (series branches in parallel) |

## 3.5 Full Adder Schematic Capture

The Full Adder was captured as a top-level gate-level schematic that instantiates the XOR, NAND, NOR and inverter symbols. The two XOR gates at the top form the Sum path. The three NAND + inverter pairs and the two NOR + inverter stages form the carry path.

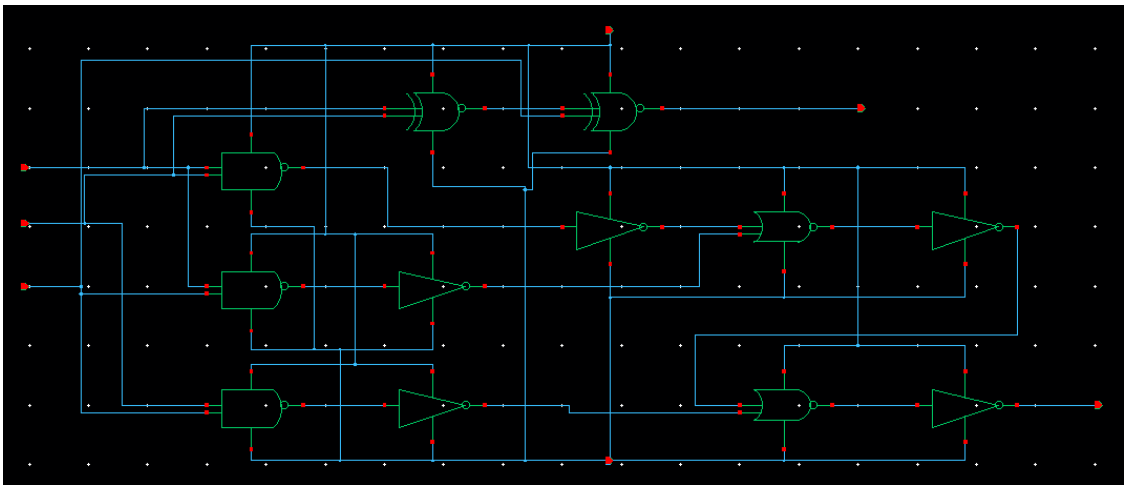


**Figure 4. Full Adder schematic captured in Cadence Virtuoso.**

## 3.6 Full Adder Test Bench and Transient Response

The Full Adder symbol was placed in a test bench with three vpulse inputs (V0 → A, V1 → B, V2 → Cin) and the vdc supply (Figure 5). The outputs are labelled Sum (pin S) and Carry (pin Cout). With periods of 20, 40 and 60 ns, the inputs apply all eight input combinations over the simulated interval.

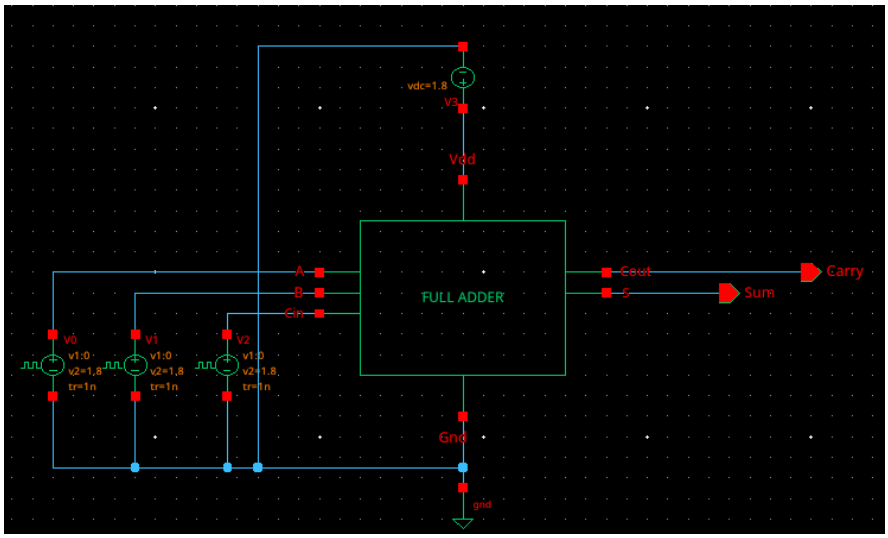



**Figure 5. Test bench for the Full Adder transient simulation.**

In the transient plot of Figure 6, net3 is A (20 ns), net2 is B (40 ns) and net1 is Cin (60 ns). Narrow spikes on Sum appear when two or more inputs change at almost the same instant (for example near 22 ns, 32 ns, 42

ns, 62 ns, 82 ns and 92 ns); they are static hazards caused by unequal path delays through the XOR and gate stages and do not change the settled logic values.

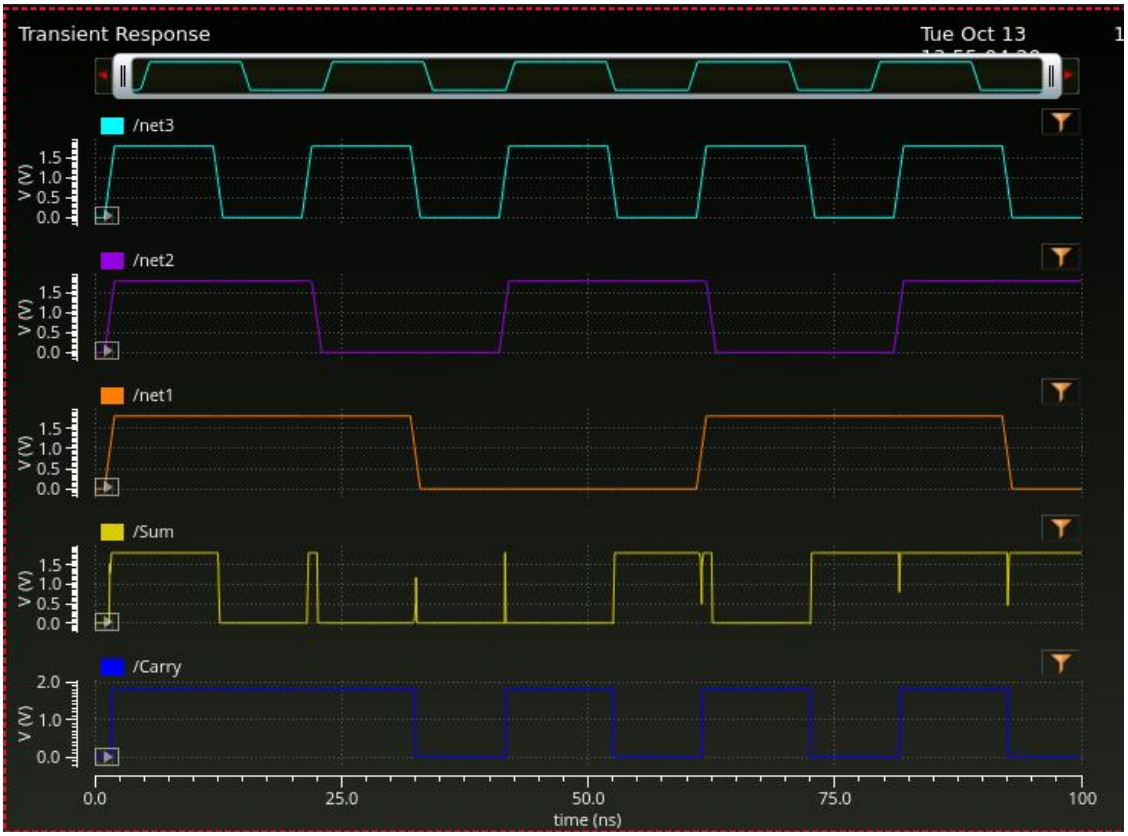



**Figure 6. Transient response of inputs A (net3), B (net2), Cin (net1) and outputs Sum and Carry.**

| Time window (ns) | A | B | Cin | Expected Sum | Expected Cout | Observed Sum / Cout |
|---|---|---|---|---|---|---|
| 2 – 12 | 1 | 1 | 1 | 1 | 1 | 1 / 1 |
| 12 – 22 | 0 | 1 | 1 | 0 | 1 | 0 / 1 |
| 22 – 32 | 1 | 0 | 1 | 0 | 1 | 0 / 1 |
| 32 – 42 | 0 | 0 | 0 | 0 | 0 | 0 / 0 |
| 42 – 52 | 1 | 1 | 0 | 0 | 1 | 0 / 1 |
| 52 – 62 | 0 | 1 | 0 | 1 | 0 | 1 / 0 |
| 62 – 72 | 1 | 0 | 1 | 0 | 1 | 0 / 1 |
| 72 – 82 | 0 | 0 | 1 | 1 | 0 | 1 / 0 |
| 82 – 92 | 1 | 1 | 1 | 1 | 1 | 1 / 1 |
| 92 – 100 | 0 | 1 | 0 | 1 | 0 | 1 / 0 |

## 3.7 Full Adder Cell Summary

| Stage | Cell(s) | Function | Inputs | Output |
|---|---|---|---|---|
| Sum 1 | XOR | P = A ⊕B | A, B | P |
| Sum 2 | XOR | Sum = P ⊕Cin | P, Cin | Sum (S) |
| Product terms | NAND + INV (×3) | A·B, A·Cin, B·Cin | (A,B), (A,Cin), (B,Cin) | T1, T2, T3 |
| Carry OR 1 | NOR + INV | T12 = T1 + T2 | T1, T2 | T12 |
| Carry OR 2 | NOR + INV | Cout = T12 + T3 | T12, T3 | Cout |

## 3.8 2×1 MUX Schematic Capture

The multiplexer was captured in Cadence Virtuoso Schematic Editor using two transmission gates (PM1/NM1 and PM2/NM2) and one select inverter (PM0/NM0).

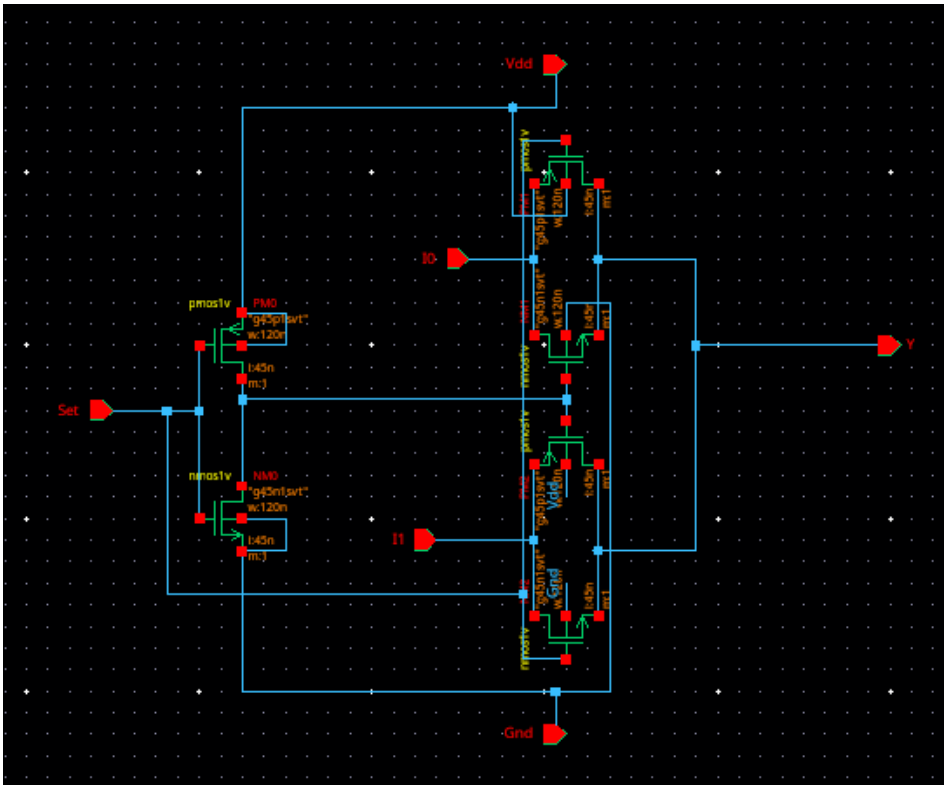



**Figure 7. 2×1 MUX schematic captured in Cadence Virtuoso.**

## 3.9 2×1 MUX Test Bench and Transient Response

The MUX symbol was placed in a test bench with three vpulse sources (V0 → I0, V1 → I1, V2 → Set) and the vdc supply (Figure 8). The output must equal I0 when S = 0 and I1 when S = 1.

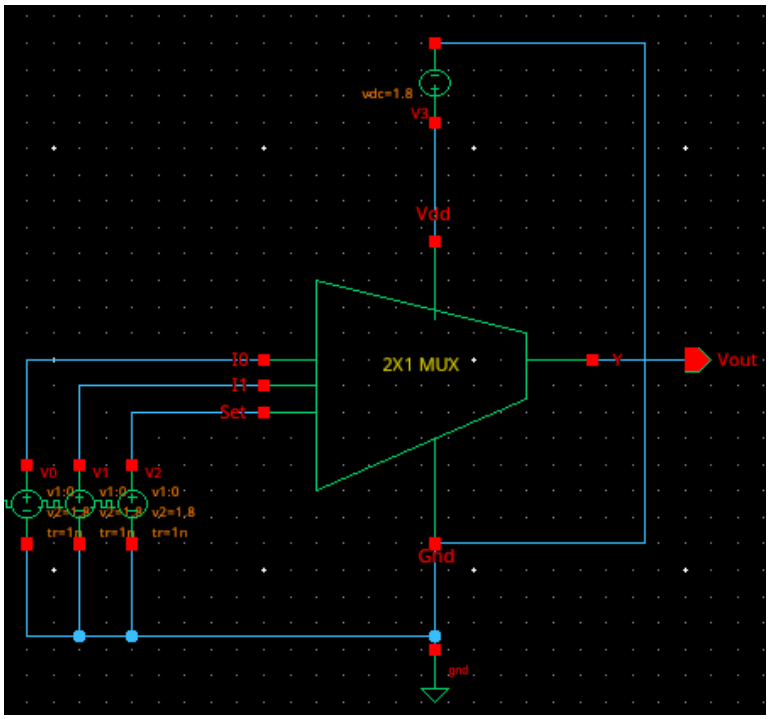



**Figure 8. Test bench for the 2×1 MUX transient simulation.**

In the transient plot of Figure 9, net3 is I0 (20 ns), net2 is I1 (40 ns) and net1 is the select input (60 ns). Small spikes at about 31 ns and 61 ns occur where the select input and a data input change together.

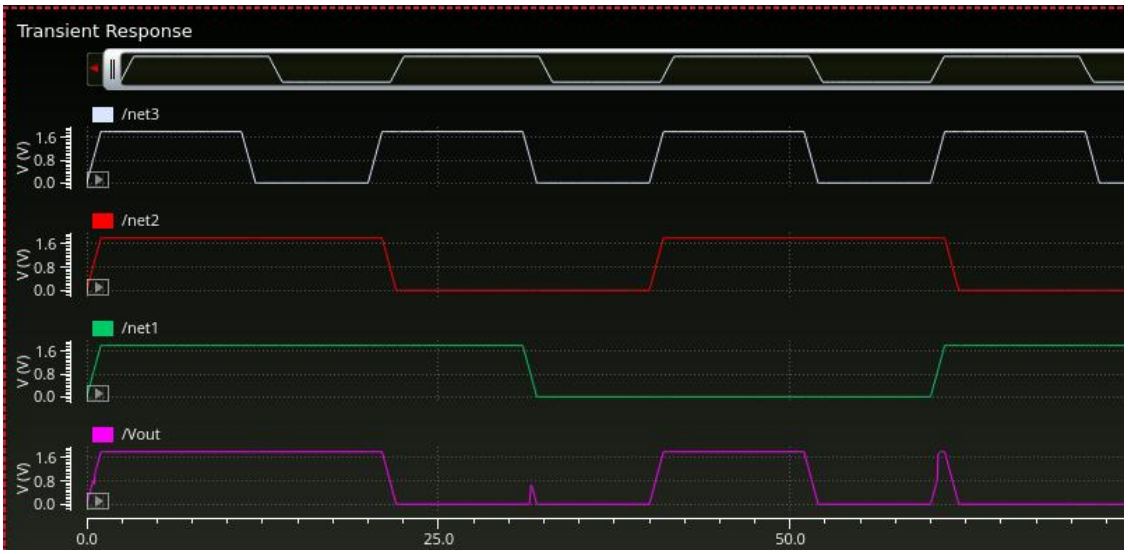



**Figure 9. Transient response of I0 (net3), I1 (net2), select (net1) and output Vout (2×1 MUX).**

| Time window (ns) | Set (S) | I0 | I1 | Expected Y | Observed Y |
|---|---|---|---|---|---|
| 2 – 12 | 1 | 1 | 1 | I1 = 1 | 1 (HIGH) |
| 12 – 22 | 1 | 0 | 1 | I1 = 1 | 1 (HIGH) |
| 22 – 32 | 1 | 1 | 0 | I1 = 0 | 0 (LOW) |
| 32 – 42 | 0 | 0 | 0 | I0 = 0 | 0 (LOW) |
| 42 – 52 | 0 | 1 | 1 | I0 = 1 | 1 (HIGH) |
| 52 – 62 | 0 | 0 | 1 | I0 = 0 | 0 (LOW) |
| 62 – 72 | 1 | 1 | 0 | I1 = 0 | 0 (LOW) |

## 3.10 2×1 MUX Device Parameter Summary

| Parameter | TG NMOS NM1, NM2 | TG PMOS PM1, PM2 | Inverter NM0 | Inverter PM0 |
|---|---|---|---|---|
| Device model | g45n1svt | g45p1svt | g45n1svt | g45p1svt |
| Width (W) | 120 nm | 120 nm | 120 nm | 120 nm |
| Length (L) | 45 nm | 45 nm | 45 nm | 45 nm |
| Multiplier (m) | 1 | 1 | 1 | 1 |
| Role | Pass device | Pass device | Generates S′ | Generates S′ |

## 3.11 Key Simulation Results

| Circuit | Logic condition | Expected / Observed |
|---|---|---|
| XOR | A B = 00 | Y = LOW |
| XOR | A B = 01 | Y = HIGH |
| XOR | A B = 10 | Y = HIGH |
| XOR | A B = 11 | Y = LOW |
| Full Adder | A B Cin = 000 | Sum = 0, Cout = 0 |
| Full Adder | A B Cin = 001 | Sum = 1, Cout = 0 |
| Full Adder | A B Cin = 010 | Sum = 1, Cout = 0 |
| Full Adder | A B Cin = 011 | Sum = 0, Cout = 1 |
| Full Adder | A B Cin = 100 | Sum = 1, Cout = 0 |
| Full Adder | A B Cin = 101 | Sum = 0, Cout = 1 |
| Full Adder | A B Cin = 110 | Sum = 0, Cout = 1 |
| Full Adder | A B Cin = 111 | Sum = 1, Cout = 1 |
| 2×1 MUX | S = 0 | Y = I0 |
| 2×1 MUX | S = 1 | Y = I1 |

# 4. Symbol Creation

## 4.1 XOR Gate Symbol Creation

After transistor-level verification, a schematic symbol was created for the XOR cell so the verified circuit could be reused as a single component in higher-level designs such as the Full Adder.

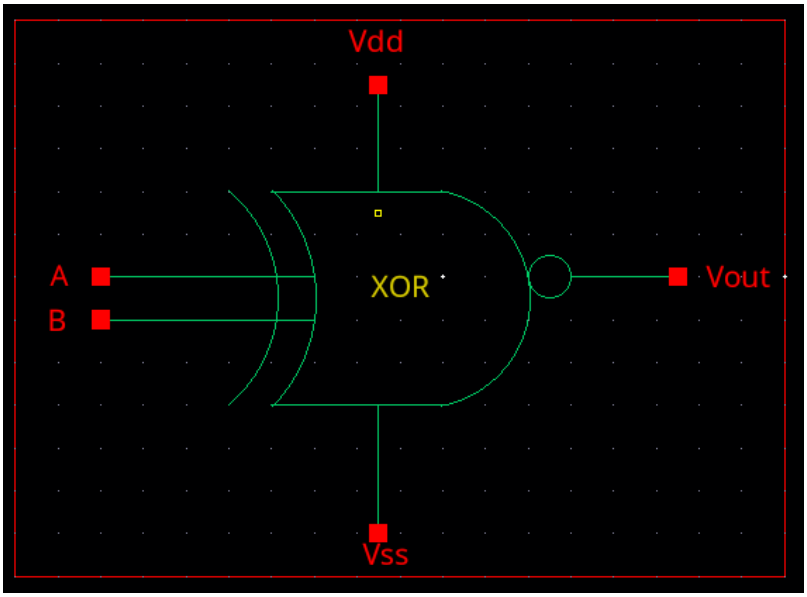



**Figure 10. Schematic symbol created for the CMOS XOR cell.**

| Pin | Direction / Type | Description |
|---|---|---|
| A | Input / Signal | First logic input |
| B | Input / Signal | Second logic input |
| Vout | Output / Signal | XOR output, Vout = A ⊕B |
| Vdd | Input / Power | Positive supply rail connected to the PMOS network |
| Vss | Input / Ground | Ground/return rail connected to the NMOS network |

## 4.2 Full Adder Symbol Creation

A reusable symbol was created for the Full Adder cell so that it can be cascaded to build multi-bit adders.

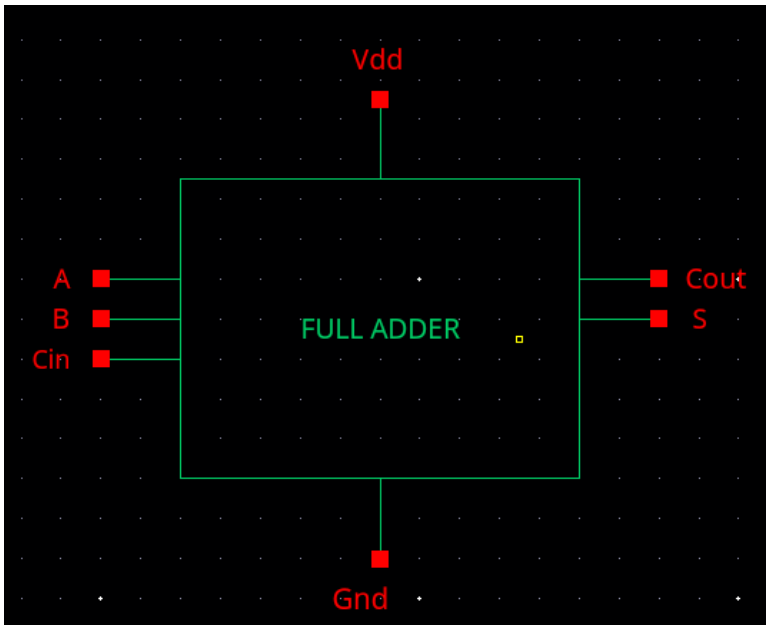



**Figure 11. Schematic symbol created for the Full Adder cell.**

| Pin | Direction / Type | Description |
|---|---|---|
| A | Input / Signal | First addend bit |
| B | Input / Signal | Second addend bit |
| Cin | Input / Signal | Carry input |
| S | Output / Signal | Sum output, S = A ⊕B ⊕Cin |
| Cout | Output / Signal | Carry output, Cout = AB + ACin + BCin |
| Vdd | Input / Power | Positive supply rail connected to the PMOS network |
| Gnd | Input / Ground | Ground/return rail connected to the NMOS network |

## 4.3 2×1 MUX Symbol Creation

After verification, a reusable symbol was created for the 2×1 MUX cell.

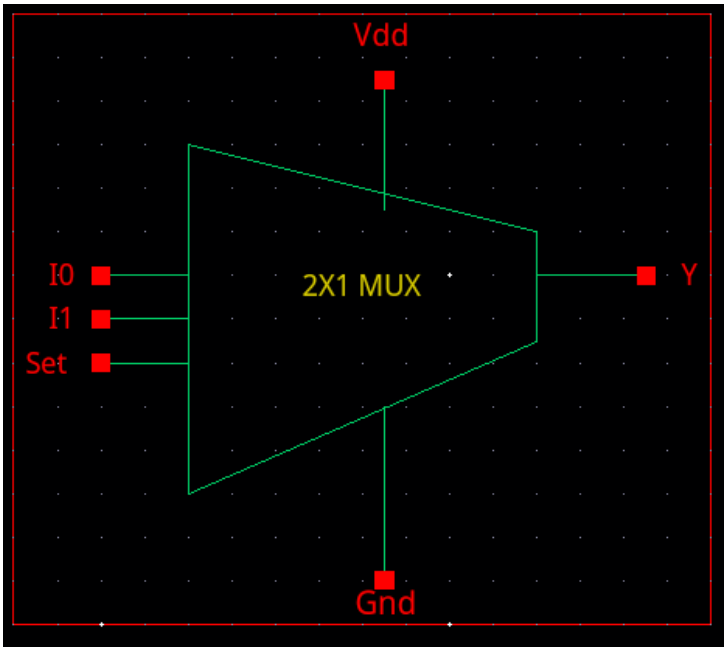



**Figure 12. Schematic symbol created for the 2×1 MUX cell.**

| Pin | Direction / Type | Description |
|---|---|---|
| I0 | Input / Signal | Data input selected when S = 0 |
| I1 | Input / Signal | Data input selected when S = 1 |
| Set | Input / Signal | Select input S |
| Y | Output / Signal | MUX output, $Y = S' \cdot I0 + S \cdot I1$ |
| Vdd | Input / Power | Positive supply rail connected to the PMOS network |
| Gnd | Input / Ground | Ground/return rail connected to the NMOS network |

# 5. Results and Discussion

## 5.1 XOR Results

- The simulated XOR output follows the expected behavior: LOW for 00 and 11, and HIGH for 01 and 10.
- The series NMOS branches pull the output LOW only when both inputs are equal.
- The series-of-parallel PMOS network pulls the output HIGH only when the inputs differ.
- Short glitches near 20–21 ns and 40 ns occur where A and B change at almost the same time; the output settles to the correct value afterwards.
- With equal 120 nm widths, the two-transistor series stacks are weaker than a single device, so the output transitions are slower than those of an inverter.

## 5.2 Full Adder Results

- The simulated Sum and Carry match the Full Adder truth table for all eight input combinations (Section 3.6).
- Sum is HIGH when an odd number of inputs are HIGH, and Cout is HIGH when at least two inputs are HIGH.
- The longest path (A or B → XOR1 → XOR2 → Sum) has the largest propagation delay. Where several inputs switch together, unequal path delays produce brief spikes on Sum before it settles.
- Hierarchical design with symbols allowed the verified cells to be reused without redrawing transistors.

## 5.3 2×1 MUX Results

- The simulated MUX output equals I0 when S = 0 and I1 when S = 1 in every observed interval.
- The transmission gates pass both logic levels without a threshold-voltage loss, which is why the MUX needs only 6 transistors.
- The output node is not driven directly from the supply rails: its drive strength depends on the on-resistance of the conducting transmission gate and on the source driving the selected input.

## 5.4 Comparison of Expected and Simulated Logic

**XOR**

| Gate | A | B | Expected Y | Simulated Y | Status |
|---|---|---|---|---|---|
| XOR | 0 | 0 | 0 | 0 | Pass |
| XOR | 0 | 1 | 1 | 1 | Pass |
| XOR | 1 | 0 | 1 | 1 | Pass |
| XOR | 1 | 1 | 0 | 0 | Pass |

**Full Adder**

| A | B | Cin | Expected Sum | Expected Cout | Simulated Sum | Simulated Cout | Status |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | Pass |
| 0 | 0 | 1 | 1 | 0 | 1 | 0 | Pass |
| 0 | 1 | 0 | 1 | 0 | 1 | 0 | Pass |
| 0 | 1 | 1 | 0 | 1 | 0 | 1 | Pass |

| A | B | Cin | Expected Sum | Expected Cout | Simulated Sum | Simulated Cout | Status |
|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 1 | 0 | 1 | 0 | Pass |
| 1 | 0 | 1 | 0 | 1 | 0 | 1 | Pass |
| 1 | 1 | 0 | 0 | 1 | 0 | 1 | Pass |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | Pass |

**2×1 MUX**

| S | I0 | I1 | Expected Y | Simulated Y | Status |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | Pass |
| 0 | 0 | 1 | 0 | 0 | Pass |
| 0 | 1 | 0 | 1 | 1 | Pass |
| 0 | 1 | 1 | 1 | 1 | Pass |
| 1 | 0 | 0 | 0 | 0 | Pass |
| 1 | 0 | 1 | 1 | 1 | Pass |
| 1 | 1 | 0 | 0 | 0 | Pass |
| 1 | 1 | 1 | 1 | 1 | Pass |

## 6. Conclusion

This experiment completed the transistor-level design and verification of a two-input static CMOS XOR gate, a one-bit Full Adder and a 2×1 multiplexer in the gpdk045 technology. The work included pre-lab theory, Boolean-function derivation, truth-table prediction, CMOS network design, transistor sizing, Cadence Virtuoso schematic capture, transient simulation with analogLib sources, and reusable symbol creation.

The XOR gate used a 12-transistor complementary topology with series-connected pull-down branches and a dual pull-up network. The Full Adder was built hierarchically at gate level from the verified XOR, NAND, NOR and inverter cells, and the multiplexer used two transmission gates and an inverter. The simulation results matched the expected truth tables for all three circuits, confirming correct logical functionality. The experiment also provided practical understanding of CMOS network topology, transmission-gate logic, device sizing, switching behavior and glitches, and hierarchical cell reuse in Cadence Virtuoso.

# Appendix: Figure Index

The figures in this report were captured directly from Cadence Virtuoso (schematic editor, ADE/VIVA waveform window and symbol editor) and inserted into the corresponding schematic, simulation and symbol sections.

| Figure | Section | Description |
|---|---|---|
| Figure 1 | 3.2 | CMOS XOR gate schematic captured in Cadence Virtuoso. |
| Figure 2 | 3.3 | Test bench for the XOR transient simulation. |
| Figure 3 | 3.3 | Transient response of inputs A (net1), B (net2) and output Vout (XOR). |
| Figure 4 | 3.5 | Full Adder schematic captured in Cadence Virtuoso. |
| Figure 5 | 3.6 | Test bench for the Full Adder transient simulation. |
| Figure 6 | 3.6 | Transient response of inputs A (net3), B (net2), Cin (net1) and outputs Sum and Carry. |
| Figure 7 | 3.8 | 2×1 MUX schematic captured in Cadence Virtuoso. |
| Figure 8 | 3.9 | Test bench for the 2×1 MUX transient simulation. |
| Figure 9 | 3.9 | Transient response of I0 (net3), I1 (net2), select (net1) and output Vout (2×1 MUX). |
| Figure 10 | 4.1 | Schematic symbol created for the CMOS XOR cell. |
| Figure 11 | 4.2 | Schematic symbol created for the Full Adder cell. |
| Figure 12 | 4.3 | Schematic symbol created for the 2×1 MUX cell. |